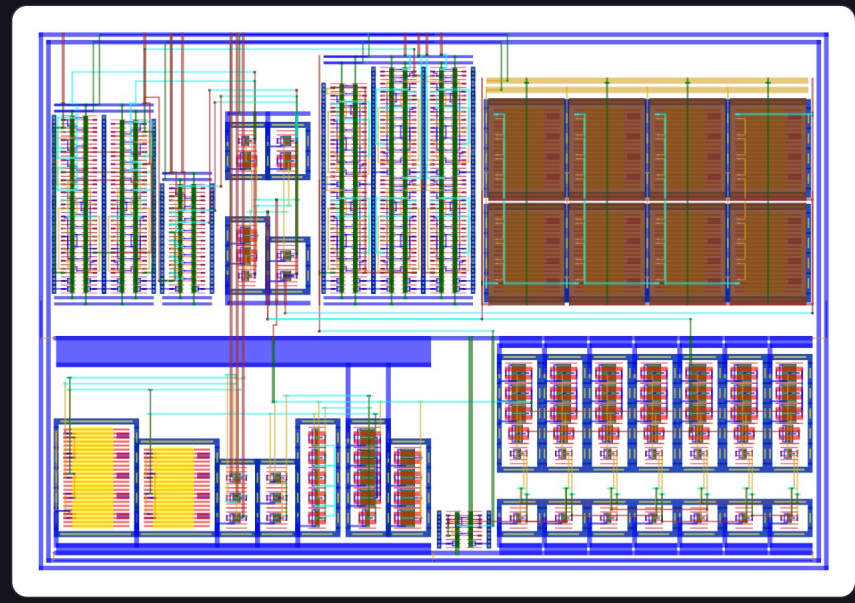
REY_PLL_SKY130A · SKYWATER 130 NM

REY_PLL
How It Was Made

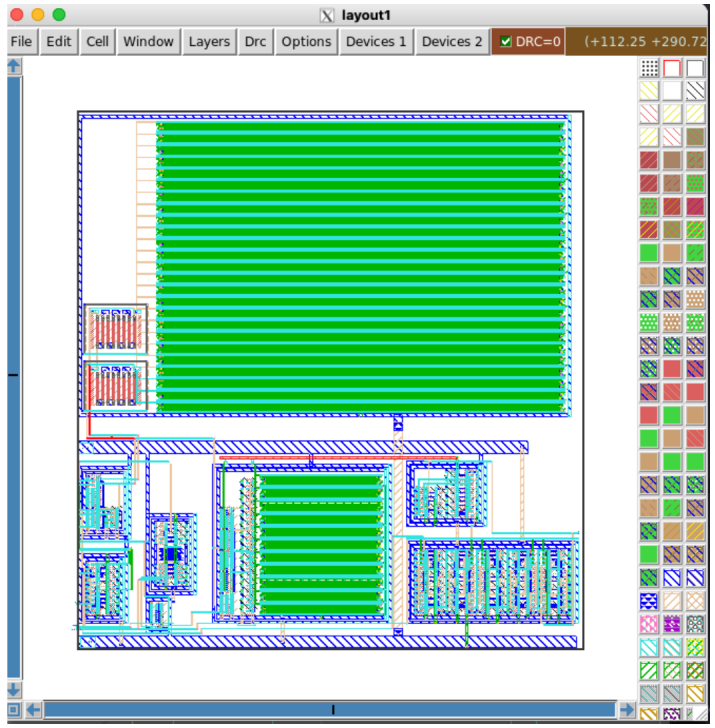
A phase-locked loop for one Tiny Tapeout tile, made by one human and one AI agent in 46 prompts.



Carsten Wulff and Claude Code · 7 to 12 November 2026

WHERE WE STARTED

sun_pll, and what was wrong with it



sun_pll_sky130nm, the layout this one starts from

A variant of sun_pll with a new oscillator

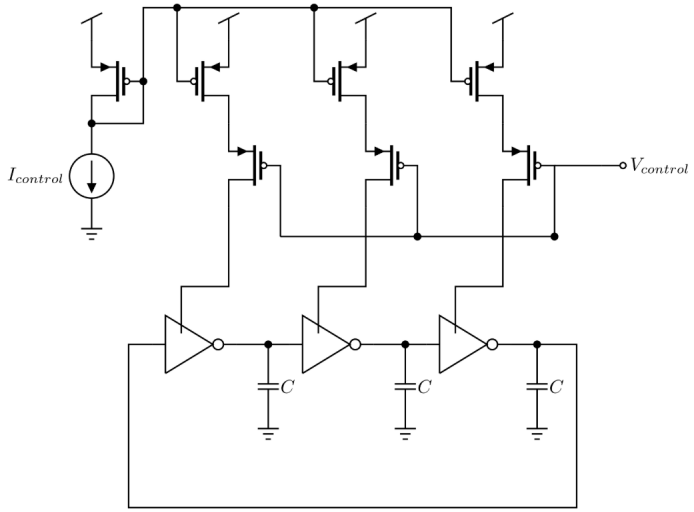
SUN_PLL	A VDD controlled ring oscillator behind an OTA buffer. Kvco about 1.6 GHz/V. It did not work in the slow corner.
REY_PLL	A current starved ring oscillator. The bias sets the frequency, VCTRL only trims it, and there is no OTA buffer.
GOAL	512 MHz from 8 MHz, over process, 1.7 to 1.9 V and −40 to 125 °C, with one trim code per die.
SIZE	One Tiny Tapeout tile, 161 × 111.52 µm.

7 NOVEMBER, EVENING

The first prompt

PROMPT 1 OF 46

```
let's start a rey_pll variant of sun_pll I want to replace the
oscillator with
https://analogicus.com/aic2027/assets/media/osc_ring_adv_tikz.svg
```



The link is a figure from the lecture notes: a current starved ring oscillator. The agent asked two things before it began.

HOW TO BUILD IT	"rey_pll_sky130a, ROSC first"
THE NEGATIVE KVCO	"Swap UP/DN at the CP"

7 TO 12 NOVEMBER 2026

The record in numbers

46

prompts, and three answers to the agent's questions

897

words in all of them together

9

of the prompts are "continue" or "go"

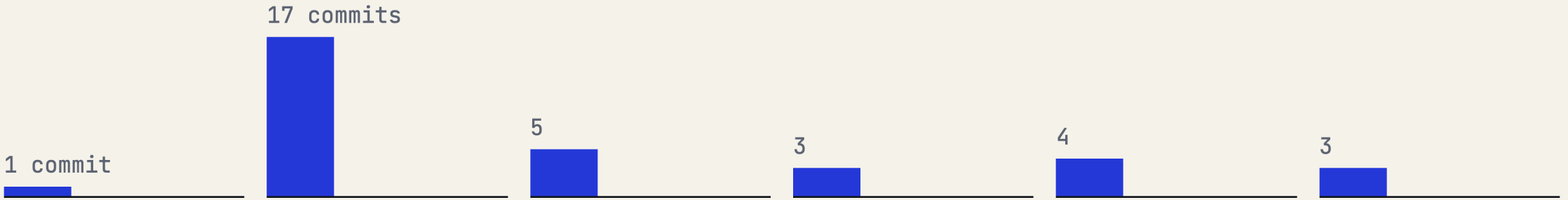
33

commits, plus 12 in cicpy and one in the shared flow

The human talked, mostly by voice and often while the agent was still working. The agent drew the schematics, wrote the generators and layout scripts, ran the simulations and checks, and wrote the documentation.

33 COMMITS IN SIX DAYS

Day by day



7 NOV

An oscillator

One sentence and a lecture figure. The oscillator and a Python model of the loop.

8 NOV

Bias, loop, first layouts

Vgs/R bias, a generator for the rest. The schematic locks at 512 MHz. The first top shorts the supplies.

9 NOV

Small cells first

A ring stage, then the bias, then the oscillator from the finished pieces. All three clean.

10 NOV

The top, too big

DRC and LVS clean at 209 × 155 µm. The tile is 161 × 111.52.

11 NOV

One tile, wrong frequency

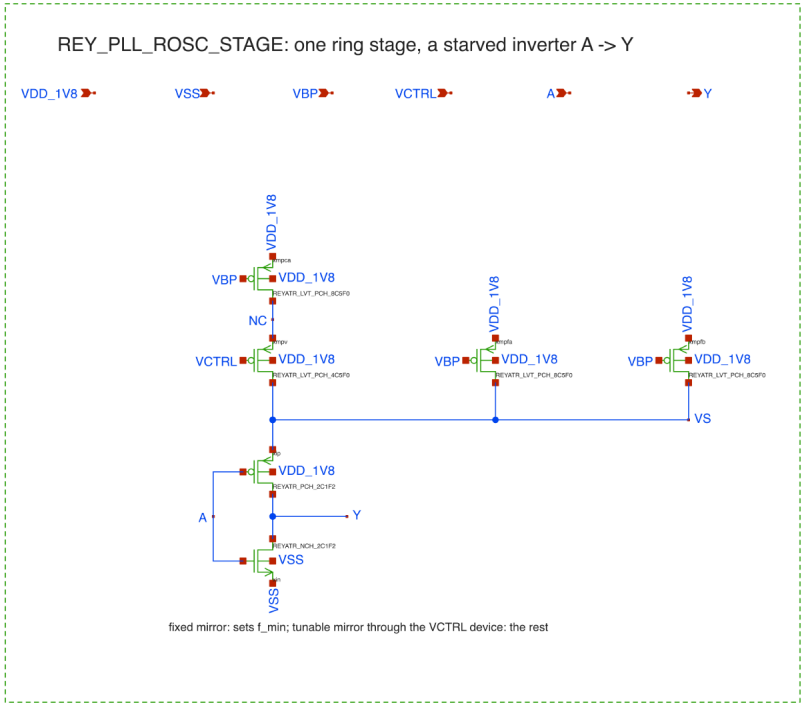
Fits one tile. Extracted, it does not reach 512 MHz. Retargeted to 128 MHz.

12 NOV

The flow

All 20 corners lock. A netgen crash in the LVS flow gets a hook in the shared flow.

The oscillator

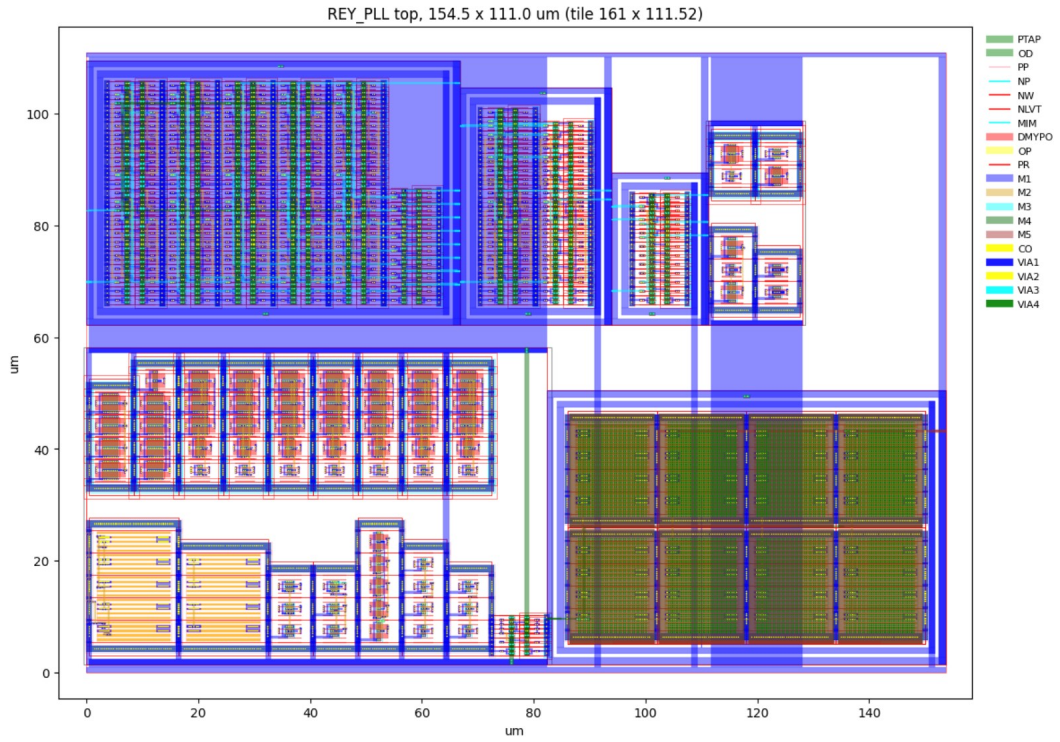


One ring stage, a starved inverter. A fixed mirror sets the minimum frequency, VCTRL trims a second one.

and let's skip the ota buffer	**No OTA buffer**
I want a prime number of stages, so 5 is fine, but also 7.	**Seven stages**
set the use the mirror to set the current, and maybe we should have a trimmed resistor or something.	**Current mirrors, 3 bit trim**
We could just make the ring oscillator transistors longer, particularly the NMOS.	**Longer transistors**
It would be nice if we can actually get it to work over temperature.	**A Vgs/R bias**

8 NOVEMBER, EVENING

The first top placement



The first placement. The supplies are one short.

Five blocks were clean by the evening. The top was not: cells and guard rings stand a little outside their boxes, so blocks placed edge to edge shorted VDD to VSS.

> PROMPT
>
> So it looks like the ground connections is what shorts everything. So you probably need to route that by hand.

> WHILE WORKING
>
> And get all the sub-blocks clean before we proceed to the top level.

9 NOVEMBER

Small cells first

FIRST PROMPT OF THE DAY

So in general, the actually maybe we have to lay out the inverter by itself first. Do that. Lay out the inverter first. General for layout, you have to combine it. Have a look at Lilo temp also. You have to combine it into groups, and then those groups has to be routed, and then further up can be routed.



1. One ring stage

8.96 × 40.60 µm

The starved inverter as its own cell, clean before anything uses it.



2. The bias

74.96 × 36.80 µm

Its own cell as well, with the split made vertical.



3. The oscillator

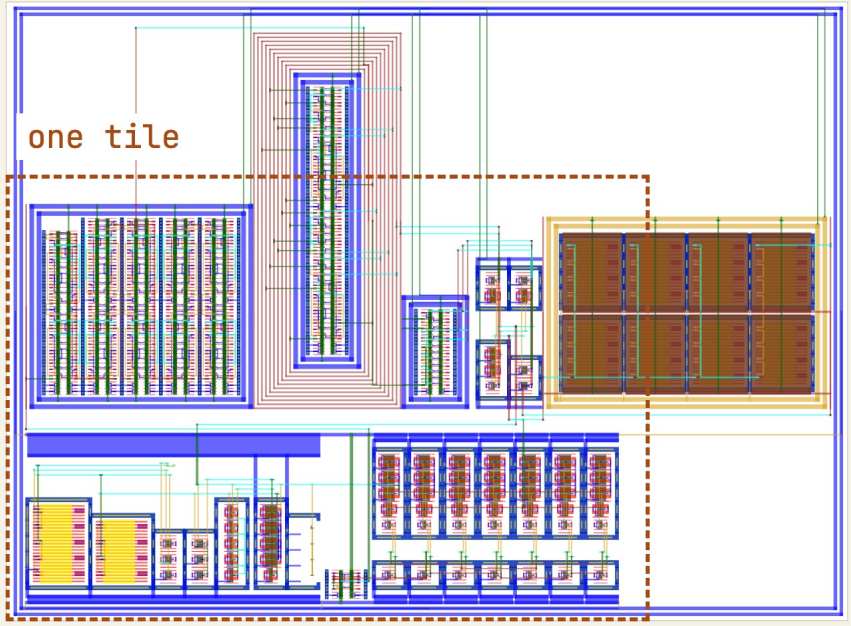
150.78 × 43.66 µm

From the finished pieces, with a top that only routes between them. Clean by the evening.

10 TO 11 NOVEMBER · BOTH DRAWN AT THE SAME SCALE

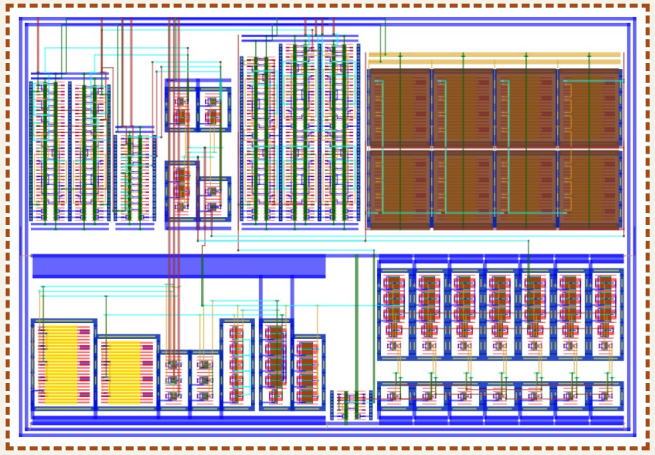
Fitting the tile

The top came together clean, and too big. The blocks alone filled 86 % of the tile before a single wire between them.



10 November

209 × 155 µm, DRC and LVS clean



11 November

156.96 × 106.94 µm, inside one tile

WHAT IT TOOK

Asked what to give up, the answer was "Squeeze, taps last".

The blocks of the top row lost their own supply rings.

The phase detector was redone as two mirrored columns.

The divider went from ten flops and eight taps to six and four.

Half of the output pins are tied low.

11 NOVEMBER · THE FIRST EXTRACTED SIMULATION

The wrong frequency

With its wiring, the oscillator is 2.4 times slower than its schematic. No trim code reaches 512 MHz.

Oscillator	Schematic	Extracted
Trim code 0	509 MHz	211 MHz
Trim code 7	953 MHz	371 MHz

About 10 fF of wiring sits on every ring node. The devices' own load is about 7 fF.

The oscillator was never simulated extracted until the top was finished. The check was cheap and the order was wrong.

THE THREE WAYS ON

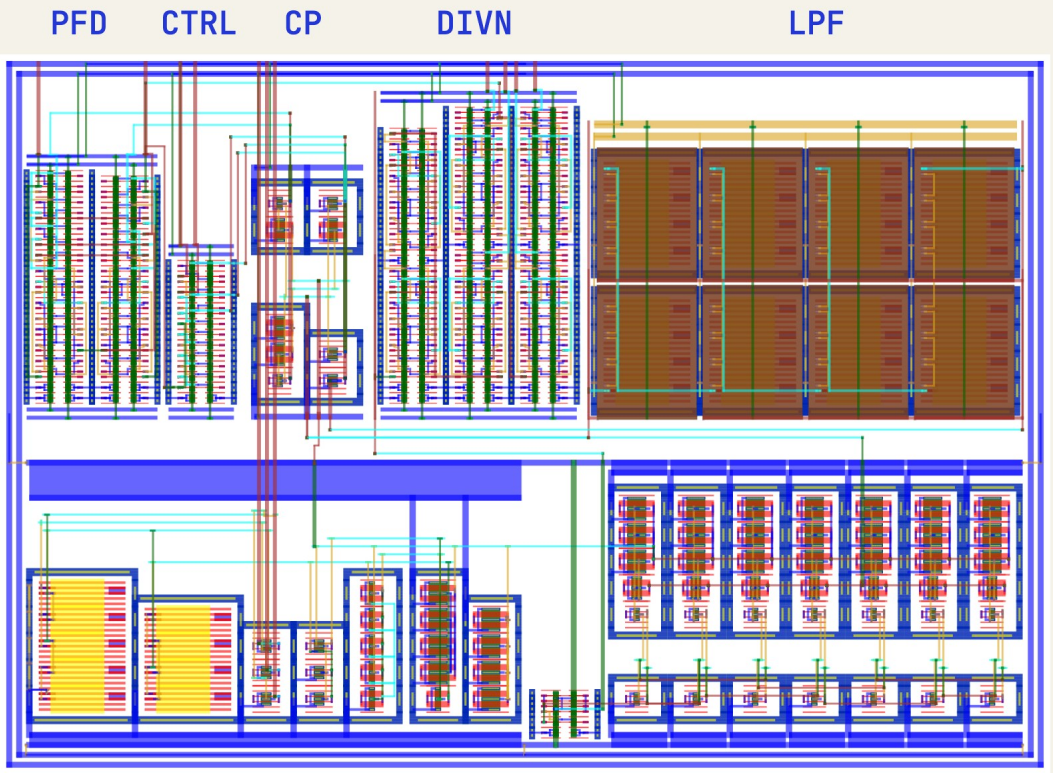
1. **128 MHz, the oscillator untouched**
2. 256 MHz, a faster oscillator
3. 512 MHz, a new oscillator

> PROMPT
>
> `Yeah, let's do option one first and get it DRC all guys clean and clean`
>
> Dictation for "DRC and LVS clean".

The loop now divides by 16 and locks at 128 MHz from the same 8 MHz reference.

WHERE WE ARE

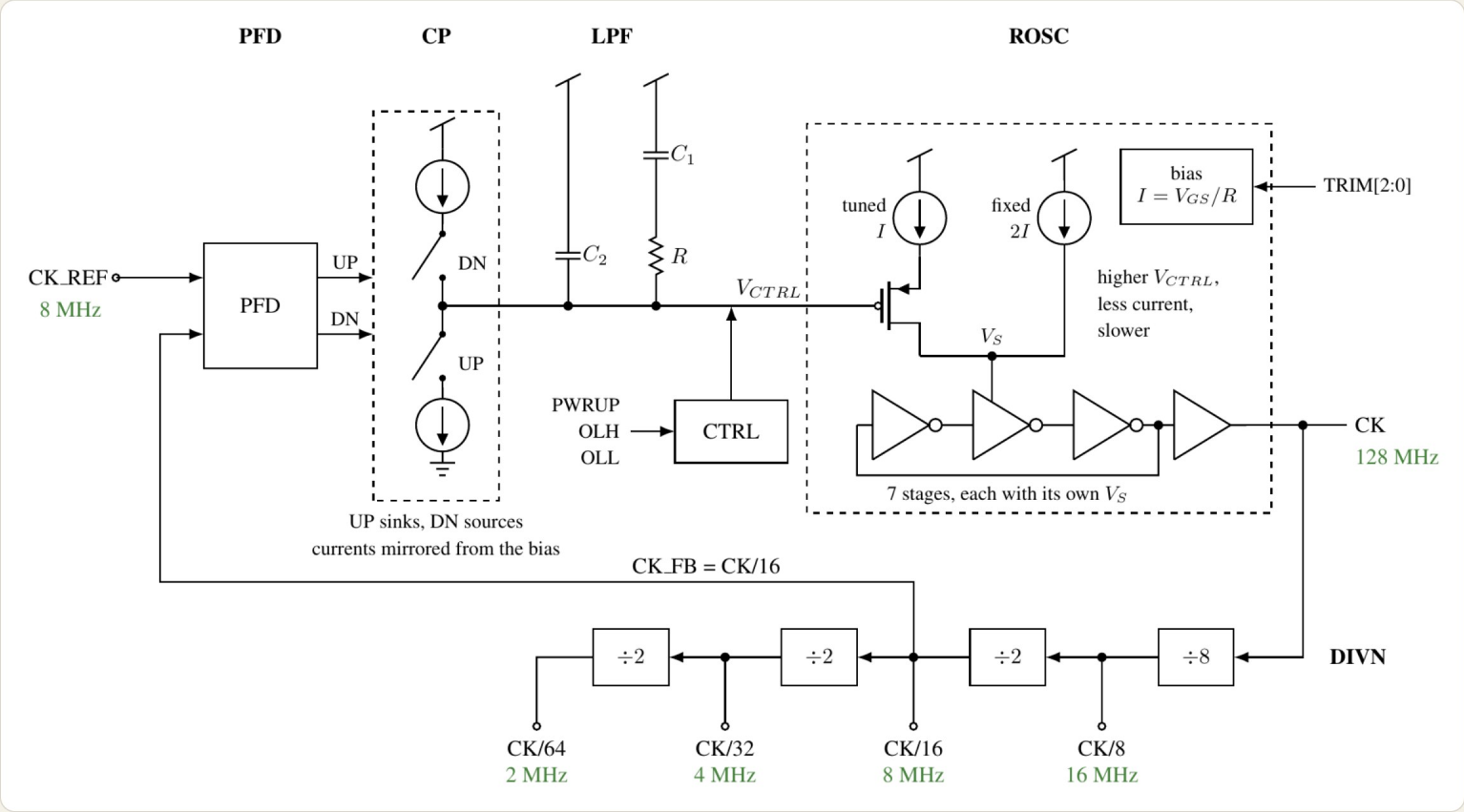
The layout



CORE	REY_PLL, 156.96 × 106.94 µm
WRAPPER	tt_um_rey_pll_wulffern, one 161 × 111.52 µm tile
CHECKS	DRC in magic and KLayout, and LVS, clean. Antenna clean on the wrapper.
GENERATED	Everything but the oscillator comes from py/gen_pll.py.
LOOP FILTER	Eight unit cells, a MiM capacitor on top of a poly resistor, 16 × 20.8 µm each.

WHERE WE ARE

The loop



128 MHz

The goal was 512 MHz.

The extracted layout locks in typical and in all 20 corners: five processes, −40 and 125 °C, 1.7 and 1.9 V.

VCTRL sits between 0.56 and 1.53 V, with ripple under 14 mV.

46 PROMPTS, EIGHT DECISIONS

What the human decided

Most of the words were permission to carry on. A few of them are why the design looks as it does.

The architecture

The lecture's current starved oscillator, no OTA buffer, a prime number of stages.

No PVT tracking

It should just work over temperature. That is where the Vgs/R bias comes from.

One tile

Said on the second morning, before any layout. It set the loop filter, the Kvco and the floorplan.

MiM over poly

The loop filter as a generated unit cell.

The layout method

No absolute coordinates, supplies by hand, small cells clean before the level above.

Fix the tool

When an automatic route made a DRC error, the fix went into cicpy.

Shared technology

Left alone for what is only this design's problem, fixed there when it is everybody's.

128 MHz

Take the frequency that works and get it clean, rather than open the oscillator again.

FIVE THINGS

What was hard

The supplies	Cells and guard rings stand a little outside their boxes, so blocks placed edge to edge shorted VDD to VSS.
The oscillator as one cell	Flat, it was not clean after a day. As a stage, a bias and a top that only routes between them, it was clean the next evening.
The tile	The blocks filled 86 % of it before a single wire between them.
The parasitics, found late	About 10 fF of wiring on every ring node, more than the devices' own load. This is the mistake of the project, and it is the agent's: the check was cheap and the order was wrong.
netgen	The comparison at the end of make lpe crashed on a ladder of poly resistors. A local setup file turns the merging off for that device.

WHERE WE ARE

Honest status

128 MHz

The goal was 512 MHz. Getting back needs about 2.4 times the current in the ring stages, or fewer stages. Not started.

The loop filter is the one sized for 512 MHz. At 128 MHz the phase margin is 20 to 38 degrees by hand calculation. The corner runs show lock, not margin.

The trim codes are from the schematic sweep. The oscillator has not been swept extracted over all corners.

Nothing is taped out and nothing is measured. The Tiny Tapeout precheck has not been run.

The netgen crash is avoided, not reported.

THE RECORD

Every prompt is on the page

ALL 46 PROMPTS	analogicus.com/rey_pll_sky130a/howitwasmade.html
THE DESIGN	github.com/wulffern/rey_pll_sky130a
THE LAYOUT TOOL	github.com/wulffern/cicpy